

3(a)	<i>any three from</i> 1 <i>differences in height show that</i> concentrations of sucrose, 0, 0.4 and 0.8 (mol dm⁻³) <i>or</i> $\leq$ / less than, 0.8 / 0.9, water moves out of Visking tubing ; A one of 0, 0.4 or 0.8 R sucrose moving 2 concentrations of sucrose 1.2, 1.6, 2.0 (mol dm⁻³) <i>or</i> $\geq$ / more than, 0.9 / 1.2, water moves into Visking tubing ; A one of 1.2, 1.6 or 2.0 R sucrose moving 3 <i>ref. to <u>net</u> water movement ;</i> <i>if water enters Visking tubing – A ora for water leaving</i> 4 external solution has <u>higher</u> water potential (than contents of Visking tubing) ; A high water potential to low water potential 5 water moves, <u>down, water potential / Ψ, gradient</u> ;	3
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Question	Answer	Marks
3(b)	<i>mark whole question to a max of four marks</i> <i>descriptions</i> 1 at concentrations, less than / $\leq$, 0.04 mol dm⁻³ all cells, burst / AW ; 2 at concentrations between 0.04 and 0.14 mol dm⁻³ decreasing percentage of cells burst / AW ; A use of percentages 3 at concentrations, greater than / $\geq$, 0.14 mol dm⁻³ no cells, burst / AW ; <i>explanations to max 3</i> 4 in low concentrations / $\leq$ 0.04, of sodium chloride water moves into cells down water potential gradient / from high Ψ to low Ψ ; 5 cells increase in, volume / size / internal pressure ; 6 <i>either</i> <u>cell membranes</u> are not strong enough to withstand increase in volume / pressure or red blood cells burst because they have no cell wall ; 7 between 0.04 and 0.14 (mol dm⁻³) water potential gradient into cells, decreases / becomes less steep / AW ; 8 above 0.14 (mol dm⁻³) / in high concentrations, water potential inside cells is the same or higher than the sodium chloride solution ; 9 at high concentration / $\geq$ 0.14, water leaves cells / cells shrink / cells shrivel / cells show crenation ;	4